Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/09/2017
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Simulation of GCR-Induced Harderian Gland and Lung Tumorigenesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	02/04/2016	**End Date:**	02/03/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mao, Jian-Hua Ph.D. (Lawrence Berkeley National Laboratory) Chang, Polly Ph.D. (SRI International)		
Grant/Contract No.:	NNJ16HP22I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Increased cancer risk remains one of the primary concerns for travel into deep space and may preclude manned missions to Mars due to the large uncertainties that currently exist in estimating cancer risk from the spectrum of radiation types found in space with available human epidemiological data on radiation-induced cancer. Existing data on human risk of cancer from exposures to X- and gamma-rays must be scaled to the many types and fluxes of radiations found in space using radiation quality factors and dose-rate modification factors. Linearity of dose-response is often assumed since the shape of the dose-response at doses below 100 mSv is unknown. However, preliminary analysis of our recently completed experiments with silicon and titanium ion suggest non-linear structure in the low-dose region for both Harderian and lung tumorigenesis. The goal of this proposal is to reduce uncertainties in the estimation of particle radiation carcinogenesis from galactic cosmic radiations (GCR). Our approach is to use the existing data on Harderian gland tumorigenesis from investigations of individual high-atomic number, and high-energy (HZE) ion beams to predict outcome of mixed ion chronic GCR exposures, and then measure Harderian gland and lung dose-dependent tumor data for such exposures. We will compare some of the measured cancer prevalence outcomes of a GCR-simulated exposure with the theoretical predictions based on experiments with individual ion beams. Our main hypothesis is that HZE carcinogenetic effects are LET- (linear energy transfer) and dose-dependent and not additive at low particle doses due to the contribution of non-targeted effects. Completion of the proposed GCR studies using a mouse strain that has already been well characterized with a low spontaneous tumor background will allow a significant test of modeling approaches used to estimate carcinogenesis risk from a simulated GCR beamline.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Completion of our research on predicting and measuring HZE tumorigenic effects to test the hypothesis that HZE carcinogenic effects and LET- and dose-dependent are not additive at low particle doses due to the contribution of non-targeted effects will allow a significant test of modeling approaches used to estimate carcinogenic risk from single-ion and mixed radiation fields. This work will also benefit estimation of risk of tumorigenesis from mixed radiation fields on Earth.
Task Progress:	• Recent experimental milestones: 1--Oct 6, 2016-Irradiated 210 CB6F1 female mice using single-or two-ion beams (within 1.5 min separation) over a 5 hr period-(p alone; p+Si; p+Fe; Si+Fe), and 60 sham-irradiated controls. These animals will be used in Harderian gland and lung tumorigenesis, chromosome aberration, and tissue archiving studies. 2--Comparing human in vitro (and ex-vivo) with murine in vivo chromosome aberration yields (in coordination with M. Hada). Various chromosome aberration experiments, e.g., with mice or with normal human blood, fibroblast, and epithelial cells, were carried out in Oct. 2016 at NASA Space Radiation Laboratory (NSRL) with the goal of linking human in vitro data to murine in vivo responses. 3--A successful 17-week post exposure harvest of murine bone marrow metaphases has been completed and is being processed for chromosome aberrations. • Reports and publications 1--Chang PY, Cucinotta FA, Bjornstad KA, Bakke J, Rosen CJ, Du N et al., Harderian Gland tumorigenesis: low-dose and LET response, Radiat Res 2016: 186:449-60. (NOTE: this publication is a product of prior funding, grant NNJ11HA94I "Harderian Gland Tumorigenesis: Low-Dose-, Low-Dose-Rate-, and LET-Response," and is listed with that project's bibliography.) 2--Siranart N, Blakely EA, Cheng A, Handa N, and Sachs RK. Mixed Beam Murine Harderian Gland tumorigenesis: Predicted dose-effect relationships if neither synergism nor antagonism occurs, Radiat Res. 2016: 186:577-591. • Presentations 1--62nd Annual Meeting of Radiation Research Society, Kona, HI. Poster, Oct 16-19, 2016. A single low dose of particle radiations results in long-term perturbation of the gut microbiome, Mao JH, Karaoz U, Lim HC, JK Bielicki JK, Bjornstad KA, Rosen CJ, Brodie EL, Chang PY, and Blakely, EA. 2--American Society for Gravitational & Space Research (ASGSR), Cleveland, OH-Symposium Presentation-Oct 27, 2016. Radiobiological Applications of Single-Beam and Mixed-Beam NSRL Capabilities. Blakely EA, Sachs RK, Chang PY, Mao JH, La Tessa C, and Rusek A. 3--International Symposium Ion Therapy (ISIT), Milan, Italy – Hot Topics Debate, Nov 3-4, 2016. Proton, C-Ions (or Else)? Blakely EA, Tommasino, F, and Scifoni E. 4--NASA Human Research Program, Predicting and Simulating GCR-Induced Tumor Risk, Blakely E, Chang P, Mao JH, Bakke J, Grover A, La Tessa C, Snyder D, Bjornstad K and Sachs R, Jan 23-26, 2017.
Bibliography Type:	Description: (Last Updated: 02/12/2026)
Abstracts for Journals and Proceedings	Blakely E, Chang P, Mao J-H, Bakke H, Gover A, Rusek A, La Tessa C, Synder E, Bjornstad K, Sachs RK. "Predicting and Simulating GCR-Induced Tumor Risk." 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017
Abstracts for Journals and Proceedings	Mao J-H, Karaoz U, Lim HC, Bielicki JK, Bjornstad KA, Rosen CJ, Brody EL, Chang PY, Blakely EA. "A single low dose of particle radiations results in long-term perturbation of the gut microbiome." Presented at the 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. , Oct-2016

Abstracts for Journals and Proceedings	Blakely EA, Sachs RK, Chang PY, Mao JH, La Tessa C, Rusek A. "Radiobiological Applications of Single-Beam and Mixed-Beam NSRL." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Blakely EA, Tommasino F, Scifoni E. "Hot Topics Debate: Proton, C-Ions (or Else)?" 2016 International Symposium on Ion Therapy, Milan, Italy, November 3-4, 2016. 2016 International Symposium on Ion Therapy, Milan, Italy, November 3-4, 2016. , Nov-2016
Articles in Peer-reviewed Journals	Siranart N, Blakely EA, Cheng A, Handa N, Sachs RK. "Mixed beam murine Harderian gland tumorigenesis: Predicted dose-effect relationships if neither synergism nor antagonism occurs." Radiation Research. 2016 Dec;186(6):577-91. Epub 2016 Nov 22. http://dx.doi.org/10.1667/RR14411.1 ; PMID: 27874325 , Dec-2016
Awards	Blakely EA. "Named a 2016 Fellow of the AAAS (American Association for the Advancement of Science) for her contributions to the fields of space-relevant and clinically-relevant particle radiobiology, February 2017." Feb-2017